Sahaj Bhadja

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Summary

Backend engineer with **2+ years of experience** building low-latency data systems, scalable schemas, and reliable production workflows across AI, analytics, and transaction-heavy platforms.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*

- Designed and optimized a **multi-tenant PostgreSQL/Azure SQL schema** supporting **500+ users** and **10K+ structured records**, reducing dashboard latency by **71%** (**1.2s to 350ms**) through indexing and query tuning.
- Built ETL-style backend workflows with **FastAPI, Supabase, SQL Server, and cloud storage**, processing **5K+ monthly jobs** with **99%+** reliability and **3x** higher throughput under concurrent load.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*

- Engineered a high-throughput grading backend handling **10K+ requests/day** using **Flask, MySQL, AWS RDS, and REST APIs**, with concurrent batch execution for **1K+** files and stable peak-load performance.
- Designed schema-aware evaluation, rubric, and cost-tracking workflows for **150+ instructors**, reducing feedback turnaround by **60%** while improving observability and operational control over model usage.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered cross-platform microservices with automated VM builds and **20+** unit/integration test suites using **Jenkins, PyTest, and Mockito**, validating releases across Windows, Linux, and macOS.
- Optimized large-scale APIs and ETL pipelines with **AWS Glue, S3, and OOP patterns**, improving response times by **30%**, pipeline efficiency by **68%**, and lowering technical debt by **20%**.

Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Built a **microservice-based resource platform** with **MERN**, WebSockets, and Azure SQL, synchronizing **2M+ records** across locations and supporting **8** operational workflows for allocation, billing, and notifications.
- Automated testing and deployment with **Jenkins** and Docker, increasing release validation to **95%+**, reducing resource wastage by **42%**, and improving lead conversion by **15%**.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*

- Built low-latency **Flask** booking and payment APIs over Firebase for a cross-platform product handling **50K+ records**, improving confirmation speed by **40%** and reducing manual refund handling by **70%**.
- Designed pricing, coupon, webhook, and analytics abstractions with **Databricks, Spark, and real-time sync**, sustaining sub-**200ms** booking flows and hourly operational reporting.

Skills

Languages: Python, Java, JavaScript/TypeScript, SQL, C++
Backend & Data: FastAPI, Flask, Node.js, Express.js, REST APIs, PostgreSQL, MySQL, SQL Server, Redis
Infrastructure: AWS, Azure, Docker, CI/CD, Jenkins, GitHub Actions, Cloud Storage, Monitoring
Systems: Microservices, Distributed Systems, ETL Pipelines, Query Optimization, Concurrency, Debugging, OOP

Projects

Big-Brain Live | *PostgreSQL, FAISS, Pinecone, REST APIs*

- Built a semantic retrieval platform over **50K+ documents** with sub-second search, metadata persistence, and incremental indexing, reducing retrieval latency by **40%** for document Q&A and summarization workflows.

FinNexus Live | *Next.js, Kubernetes, MongoDB, WebSockets*

- Engineered a real-time fintech platform for **50K+ users** and **1M+ transactions**, building secure transaction workflows, low-latency synchronization, and scalable release infrastructure for high-volume event streams.

Chat2SQL GitHub | *Java, TypeScript, Docker, PostgreSQL*

- Built an AI-powered SQL workspace supporting **10+** databases, enabling schema exploration, SQL generation, and structured query execution from a single interface across Java and TypeScript services.